

BRIAN PRENDES

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EDUCATION

Yale University

Bachelor of Science in Physics | QuestBridge Scholar

New Haven, CT
Expected May 2029

- Relevant Coursework: Classical Mechanics (PHY 2000), Electromagnetism (PHY 181), Linear Algebra (MATH 2220), Computing for Engineers (ENAS 1300), Engineering Innovation & Design (ENAS 1180), Information Systems (CPSC 2000), Mechanical Design (MENG 1105).
- Activities: Yale Undergraduate Quantum Computing (YUQC), Research Assistant, Engineers Without Borders, ASME, Y-AID.

Miami Dade College

Associate of Arts | GPA: 4.0/4.0 | Dual-enrollment cohort; mentored peers on registration and engagement.

Miami, FL
July 2025

EXPERIENCE & RESEARCH

Yale University — Quantum Optics & Thermal Management

Undergraduate Research Assistant

New Haven, CT
Present

- Engineered thermal management solutions for a Spontaneous Parametric Down-Conversion (SPDC) setup generating entangled photon pairs.
- Optimized a custom copper-sandwich enclosure for a non-linear crystal to mitigate extreme thermal gradients.
- Integrated high-speed, low-thermal-mass electronic sensors and developed mathematical models to empirically calibrate internal crystal temperature based on asymmetric external heat distribution.

Yale University — Materials Science

Research Assistant

New Haven, CT
Aug 2025 – Jan 2026

- Investigate phase transitions of amorphous metals and salts for structural applications using thermal molding.
- Conduct thermal experiments on metal wire formation to characterize Plateau-Rayleigh instability patterns.
- Manage experimental setups and perform rigorous data analysis on material deformation metrics.

ASME at Yale

Program Chair

New Haven, CT
Nov 2025 – Present

- Coordinate technical agendas for meetings; organize guest speaker series and career fairs for engineering students.
- Facilitate networking opportunities connecting mechanical engineering students with industry leaders.

First Year Scholars at Yale (FSY) & Math Honor Society

Social Impact Data Analyst | Math Tutor & Treasurer

Miami, FL / New Haven, CT
2022 – July 2025

- Collected and verified gun violence data for municipal databases (FSY); tutored Algebra through AP Calc BC; managed society funds as Treasurer.

TECHNICAL & ENGINEERING PROJECTS

MIT iQuHack — Quantum Computing Hackathon | IBM Qiskit, CUDA Quantum QPP, IQM Emerald

Jan 2026 – Feb 2026

- Benchmarked limits of Genuine Multipartite Entanglement (GME) on the 54-qubit IQM Emerald processor by implementing a Loophole-Free Stabilizer Witness to isolate true quantum signatures.
- Characterized hardware noise via Ramsey Interferometry and mapped energy decay limits, modeling hardware performance against a Discrete Depolarizing Noise Channel using CUDA Quantum QPP.
- Minimized circuit depth for linear hardware topology by decomposing complex algorithms into native CZ gates via Clifford Conjugation.

Wheelchair Scale Project | Yale University

Feb 2026 – Present

- Designing an accessible, cost-effective scale system utilizing everyday materials to enable wheelchair users to independently and safely measure their own weight.

3D Printed Prosthetic Arm | Onshape, FDM 3D Printing

Feb 2026 – Present

- Fabricating a functional 3D-printed prosthetic arm for a pediatric patient to restore mobility and object manipulation, significantly reducing financial burdens for the patient's family.

V.E.G.A. — The Gamified Gripper Assistant | Arduino, C++, Onshape, Sensors

- Designed a humanoid chassis in Onshape with potentiometers for grip measurement; programmed an Arduino with endurance/repetition game modes and Morse Code learning.
- Engineered a force-sensitive navigation system allowing users to select menu options via grip pressure.

Pong Game Console | Circuit Design, C++, 3D Printing

- Engineered a retro-style enclosure via Onshape/3D printing; soldered Arduino Nano + OLED display onto custom protoboard; programmed game physics and hardware drivers in C++.

Rehabilitation Glove | Yale Assistive Innovation and Design (Y-AID)

- Member of CAD design and fabrication team for a silicone rehabilitation glove to aid physical therapy.

SKILLS & INTERESTS

Software: Python, C++, Processing, Onshape (CAD), IBM Qiskit, CUDA Quantum QPP, Excel **Hardware:** IQM Emerald (54-qubit), Arduino, Circuit Design, Soldering, 3D Printing, Sensors

Research: Quantum Optics, GME, Thermal Analysis, Phase Transitions **Languages:** English (Native), Spanish (Native)